G10Adv	Mathematics	الرياضيات	T1 · Practice

Answer key	مفتاح الإجابة
Answer key · solutions	مفتاح الإجابة والحلول

1 · A	2 · B	3 · A	4 · A	5 · A
6 · D	7 · D	8 · A	9 · A	10 · B
11 · A	12 · B	13 · A	14 · B	15 · B
16 · B	17 · B	18 · B	19 · A	20 · B
21 · B	22 · A	23 · A	24 · B	25 · A
26 · A	27 · A	28 · A	29 · A	30 · A
31 · A	32 · B	33 · B	34 · A	35 · A
36 · A	37 · A	38 · B	39 · B	40 · A
41 · A				

Answer key · solutions | الحلول

Answer key · solutions | **مفتاح الإجابة والحلول**

1 · Practice · KBZ eSolutions manual · PDF p.122 · answer A

Simplify. $(3c^3 - c + 11) - (c^2 + 2c + 8)$

1. $(3c^3 - c + 11) + (-c^2 - 2c - 8)$ distribute the minus sign
2. $3c^3 - c^2 + (-c - 2c) + (11 - 8)$ group like terms
3. $\boxed{3c^3 - c^2 - 3c + 3}$

2 · Practice · KBZ eSolutions manual · PDF p.122 · answer B

Simplify. $(2x - 2y + 1) - (3y + 4x)$

1. $(2x - 2y + 1) + (-3y - 4x)$ distribute the minus sign
2. $(2x - 4x) + (-2y - 3y) + 1$ group like terms
3. $\boxed{-2x - 5y + 1}$

3 · Practice · KBZ eSolutions manual · PDF p.126 · answer A

Find the product. $(2y - 11)(y^2 - 3y + 2)$

1. $2y(y^2 - 3y + 2) - 11(y^2 - 3y + 2)$ distributive property
2. $2y^3 - 6y^2 + 4y - 11y^2 + 33y - 22$ multiply
3. $\boxed{2y^3 - 17y^2 + 37y - 22}$ combine like terms

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

4 · Practice · KBZ eSolutions manual · PDF p.126 · answer A

Find the product. $(4a+7)(9a^2+2a-7)$

1. $4a(9a^2+2a-7)+7(9a^2+2a-7)$ distributive property
2. $36a^3+8a^2-28a+63a^2+14a-49$ multiply
3. $\boxed{36a^3+71a^2-14a-49}$ combine like terms

5 · Practice · KBZ eSolutions manual · PDF p.134 · answer A

Factor the polynomial, if possible. $5x^2+34x+24$

1. $ac=5\cdot 24=120:$ 4 and 30 add to 34 find m and p
2. $5x^2+30x+4x+24=5x(x+6)+4(x+6)$ group
3. $\boxed{(5x+4)(x+6)}$

6 · Practice · KBZ eSolutions manual · PDF p.140 · answer D

Factor the polynomial, if possible. $2x^2+3x+6$

1. $ac=12:$ factor pairs $(1,12),(2,6),(3,4)$
2. sums $13,\ 8,\ 7 \neq 3$ none adds to b = 3
3. $\boxed{\text{prime}}$ cannot be factored with integers

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

7 · Practice · KBZ eSolutions manual · PDF p.140 · answer D

Factor the polynomial, if possible. $5x^2 + 3x + 4$

1. $ac = 20:\ (1,20), (2,10), (4,5)$
2. sums $21,\ 12,\ 9$ ⊟ 3
3. $\boxed{\text{prime}}$

8 · Practice · KBZ eSolutions manual · PDF p.166 · answer A

Find a value for m that makes $4x^4 - 44x^2 + m$ a perfect square trinomial, then factor it.

1. $2(2x^2)(n) = -44x^2 \Rightarrow n = -11$ middle term = 2·(first)·(second)
2. $m = n^2 = (-11)^2 = 121$
3. $\boxed{4x^4 - 44x^2 + 121 = (2x^2 - 11)^2}$

9 · Practice · KBZ eSolutions manual · PDF p.2 · answer A

In quadrilateral QRST the interior angles are $Q = (2x + 5)°,\ R = x°,\ S = (2x + 7)°,\ T = x°$. Find the measure of each angle.

1. $\text{sum} = (4 - 2) \cdot 180° = 360°$ polygon interior angle sum
2. $(2x + 5) + x + (2x + 7) + x = 360 \Rightarrow 6x + 12 = 360 \Rightarrow x = 58$
3. $\boxed{Q = 2(58) + 5 = 121°,\ S = 2(58) + 7 = 123°}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

10 · Practice · KBZ eSolutions manual · PDF p.9 · answer B

Quadrilateral ABCD is a rectangle. If $m\angle ADB = (4x + 8)^\circ$ and $m\angle DBA = (6x + 12)^\circ$, find x.

1. $\angle DAB = 90^\circ$ (rectangle)
2. $(4x + 8) + (6x + 12) + 90 = 180$ triangle angle sum
3. $\boxed{10x + 110 = 180 \Rightarrow x = 7}$

11 · Practice · KBZ eSolutions manual · PDF p.6 · answer A

In a quadrilateral, one pair of opposite sides is both parallel and congruent. Which statement is correct?

1. $\boxed{\text{One pair of opposite sides parallel and congruent} \Rightarrow \text{parallelogram}}$ test for parallelograms

12 · Practice · KBZ eSolutions manual · PDF p.6 · answer B

Both pairs of opposite angles of a quadrilateral are congruent. Is it a parallelogram?

1. $\boxed{\text{Both pairs of opposite angles congruent} \Rightarrow \text{parallelogram}}$ test for parallelograms

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

13 · Practice · KBZ eSolutions manual · PDF p.15 · answer A

In trapezoid ABCD, $\overline{AD} \parallel \overline{BC}$ with $A(-3,2),\ B(-3,-3),\ C(3,0),\ D(1,4)$. Find the endpoints of the midsegment.

1. midsegment joins the midpoints of the legs AB and CD
2. $M_{AB} = \left(\frac{-3-3}{2}, \frac{2-3}{2}\right) = (-3, -0.5)$
3. $\boxed{M_{CD} = \left(\frac{3+1}{2}, \frac{0+4}{2}\right) = (2,2)}$

14 · Practice · KBZ eSolutions manual · PDF p.17 · answer B

Triangle J'K'P' is the image of JKP. $J(2,4),\ K(4,4)$ and $J'(4,8),\ K'(8,8)$. Find the scale factor.

1. $JK = \sqrt{(4-4)^2 + (4-2)^2} = 2$
2. $J'K' = \sqrt{(8-8)^2 + (8-4)^2} = 4$
3. $\boxed{\frac{J'K'}{JK} = \frac{4}{2} = 2}$ image over original

15 · Practice · KBZ eSolutions manual · PDF p.24 · answer B

Two polygons are similar. Corresponding sides give $\frac{x+1}{8} = \frac{3x+1}{20}$. Find x.

1. $20(x+1) = 8(3x+1)$ cross products
2. $20x + 20 = 24x + 8$
3. $\boxed{12 = 4x \Rightarrow x = 3}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

16 · Practice · Best Math 2024-25 solutions · Q13 · answer B

A'B'C'D' is the image of ABCD under a dilation. $C(6,-6)$, $D(-6,-6)$ and $C'(4,-4)$, $D'(-4,-4)$. Find the scale factor.

1. $CD = \sqrt{(-6-6)^2+0^2} = 12$
2. $C'D' = \sqrt{(-4-4)^2+0^2} = 8$
3. $\boxed{\frac{C'D'}{CD} = \frac{8}{12} = \frac{2}{3}}$ image over original

17 · Practice · Best Math 2024-25 solutions · Q13 · answer B

A figure and its dilated image are drawn on the coordinate plane. Two matching points: original $(-2,0)$ and $(-2,3)$; image $(-5,0)$ and $(-5,7.5)$. Find the scale factor.

1. original height $= 3-0 = 3$
2. image height $= 7.5-0 = 7.5$
3. $\boxed{\frac{7.5}{3} = 2.5}$ image over original

18 · Practice · Best Math 2024-25 solutions · Q13 · answer B

A dilation with center the origin and scale factor $k = 0.5$ is applied. Find the image of $K(-4,4)$.

1. $(x,y) \to (kx,ky)$ dilation rule
2. $\boxed{K'(0.5\cdot(-4),\ 0.5\cdot 4) = (-2,2)}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

19 · Practice · KBZ eSolutions manual · PDF p.30 · answer A

In △RST, ∠S = 42° and ∠T = 98°. In △UVW, ∠V = 98° and ∠U = 40°. Are the triangles similar?

1. $m\angle R = 180^\circ - (98^\circ + 42^\circ) = 40^\circ$ triangle angle sum
2. $\boxed{\angle T \cong \angle V,\ \angle R \cong \angle U \Rightarrow \triangle RST \sim \triangle UWV}$ AA similarity

20 · Practice · KBZ eSolutions manual · PDF p.39 · answer B

In △ADC, point E lies on AD and point B lies on AC with $\overline{EB} \parallel \overline{DC}$. If AB = 6, BC = 4 and AE = 9, find ED.

1. $\frac{AE}{ED} = \frac{AB}{BC}$ triangle proportionality theorem
2. $\frac{9}{ED} = \frac{6}{4}$
3. $\boxed{36 = 6\,ED \Rightarrow ED = 6}$ cross products

21 · Practice · KBZ eSolutions manual · PDF p.39 · answer B

In △ADC, point E lies on AD and point B lies on AC with $\overline{EB} \parallel \overline{DC}$. If AB = 12, AC = 16 and ED = 5, find AE.

1. $\frac{12}{16} = \frac{AE}{AE + 5}$
2. $12AE + 60 = 16AE$ cross products
3. $\boxed{4AE = 60 \Rightarrow AE = 15}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

22 · Practice · KBZ eSolutions manual · PDF p.53 · answer A

An angle bisector of a triangle divides the opposite side into 8 and 12. The sides next to those parts are m and 26. Find m to the nearest tenth.

1. $\frac{12}{8} = \frac{26}{m}$ triangle angle bisector theorem
2. $12m = 208$ cross products
3. $\boxed{m \approx 17.3}$

23 · Practice · KBZ eSolutions manual · PDF p.53 · answer A

An angle bisector divides a side into 7 and x. The sides next to those parts are 20 and 24 (7 goes with 20). Find x.

1. $\frac{7}{x} = \frac{20}{24}$ angle bisector theorem
2. $168 = 20x$
3. $\boxed{x = 8.4}$

24 · Practice · KBZ eSolutions manual · PDF p.54 · answer B

Find the geometric mean between 4 and 6.

1. $x = \sqrt{ab}$ definition
2. $\boxed{x = \sqrt{4 \cdot 6} = \sqrt{24} = 2\sqrt{6} \approx 4.9}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

25 · Practice · KBZ eSolutions manual · PDF p.54 · answer A

Find the geometric mean between $\frac{1}{2}$ and 2.

1. $x = \sqrt{ab} = \sqrt{\frac{1}{2} \cdot 2}$
2. $\boxed{= \sqrt{1} = 1}$

26 · Practice · KBZ eSolutions manual · PDF p.57 · answer A

The legs of a right triangle are 3 and 3. Find the hypotenuse x.

1. $a^2 + b^2 = c^2$
2. $3^2 + 3^2 = x^2 \Rightarrow 18 = x^2$
3. $\boxed{x = \sqrt{18} = 3\sqrt{2} \approx 4.2}$

27 · Practice · KBZ eSolutions manual · PDF p.74 · answer A

A square has diagonal 13. Find the side length x and the angle y that the diagonal makes with a side.

1. a diagonal splits the square into two $45°$-$45°$-$90°$ triangles
2. $h = \ell\sqrt{2} \Rightarrow 13 = x\sqrt{2}$
3. $\boxed{x = \frac{13}{\sqrt{2}} = \frac{13\sqrt{2}}{2},\ y = 45°}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

28 · Practice · KBZ eSolutions manual · PDF p.80 · answer A

In a right triangle the hypotenuse is 40 and the side opposite angle y is 22. Find y to the nearest tenth.

1. $\sin y^\circ = \dfrac{\text{opp}}{\text{hyp}} = \dfrac{22}{40}$
2. $\boxed{y = \sin^{-1}\left(\dfrac{22}{40}\right) \approx 33.4}$ calculator in degree mode

29 · Practice · KBZ eSolutions manual · PDF p.88 · answer A

Carlos (6 ft tall) sights the top of a garage at a 42° angle of elevation. He steps back 20 ft and the angle is 10°. How tall is the garage to the nearest foot?

1. $\tan 42^\circ = \dfrac{y}{x},\ \tan 10^\circ = \dfrac{y}{x+20}$ two right triangles
2. $0.9x = 0.176x + 3.5 \Rightarrow x \approx 4.8$ equal y
3. $\boxed{y = 4.8 \tan 42^\circ \approx 4.3,\ 6 + 4.3 \approx 10 \text{ ft}}$ add his height

30 · Practice · Best Math 2024-25 solutions · Q5 · answer A

A triangular piece of scenery has two sides of 8 ft and 10 ft with an included angle of 34°. Find its area to the nearest tenth.

1. $\text{Area} = \dfrac{1}{2}ab\sin C$ two sides and the included angle
2. $= \dfrac{1}{2}(8)(10)\sin 34^\circ$
3. $\boxed{\approx 22.4 \text{ ft}^2}$ calculator in degree mode

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

31 · Practice · KBZ eSolutions manual · PDF p.154 · answer A

Factor the expression. $9a^2 - 4b^2$

1. $9a^2 - 4b^2 = (3a)^2 - (2b)^2$ write in the form a² − b²
2. $\boxed{(3a + 2b)(3a - 2b)}$ factor the difference of squares

32 · Practice · KBZ eSolutions manual · PDF p.154 · answer B

Factor the expression. $16 - x^2$

1. $16 - x^2 = (4)^2 - (x)^2$
2. $\boxed{(4 + x)(4 - x)}$ difference of squares

33 · Practice · Best Math 2024-25 solutions · Q10 · answer B

The exterior angles of a hexagon measure $x°,\ 2x°,\ (x + 10)°,\ (x + 18)°,\ 3x°,\ (x - 1)°$. Find x.

1. sum of the exterior angles $= 360°$ for any convex polygon
2. $9x + 27 = 360$ add the six angles
3. $\boxed{9x = 333 \Rightarrow x = 37}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

34 · Practice · Best Math 2024-25 solutions · Q10 · answer A

The exterior angles of a quadrilateral measure $(x+10)°,\ 31°,\ (2x-42)°,\ (x-11)°$. Find x.

1. $(x+10)+31+(2x-42)+(x-11)=360$ exterior angles add to 360°
2. $4x-12=360$
3. $\boxed{4x=372 \Rightarrow x=93}$

35 · Practice · KBZ eSolutions manual · PDF p.34 · answer A

ROOFING. In the roof outline △DEF ~ △FBC by SAS with DF = 10 ft, FC = 16 ft and FB = 8 ft. Find DE = x.

1. $\frac{DF}{FC}=\frac{DE}{FB}$ similar polygons
2. $\frac{10}{16}=\frac{x}{8}$
3. $\boxed{80=16x \Rightarrow x=5 \text{ ft}}$

36 · Practice · KBZ eSolutions manual · PDF p.34 · answer A

RADIO. At the same time an 8-foot shadow is cast by a radio tower, a yardstick (36 in) casts a shadow 0.5 in long. How tall is the tower?

1. $\frac{h}{8 \text{ ft}}=\frac{36 \text{ in}}{0.5 \text{ in}}$ same units on each side
2. $\frac{36}{0.5}=72$
3. $\boxed{h=8(72)=576 \text{ ft}}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

37 · Practice · Best Math 2024-25 solutions · Q15 · answer A

Find the distance between $Y(-1, 0)$ and $Z(0, -1)$.

1. $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. $\boxed{YZ = \sqrt{(0 - (-1))^2 + (-1 - 0)^2} = \sqrt{1 + 1} = \sqrt{2}}$

38 · Practice · Best Math 2024-25 solutions · Q15 · answer B

Find the distance between $X(-3, -2)$ and $Y(-1, 0)$.

1. $XY = \sqrt{(-1 - (-3))^2 + (0 - (-2))^2}$
2. $\boxed{= \sqrt{4 + 4} = \sqrt{8} \approx 2.83}$

39 · Practice · Best Math 2024-25 solutions · Q15 · answer B

Find the distance between $X(-3, -2)$ and $Z(0, -1)$.

1. $XZ = \sqrt{(0 - (-3))^2 + (-1 - (-2))^2}$
2. $\boxed{= \sqrt{9 + 1} = \sqrt{10} \approx 3.16}$

Answer key · solutions	الحلول
Answer key · solutions	مفتاح الإجابة والحلول

40 · Practice · KBZ eSolutions manual · PDF p.96 · answer A

In △ABC, ∠A = 15°, ∠C = 72° and AB = c = 375. Find BC = x to the nearest tenth.

1. $\frac{\sin A}{a} = \frac{\sin C}{c}$ law of sines
2. $\frac{\sin 15^\circ}{x} = \frac{\sin 72^\circ}{375}$
3. $\boxed{x = \frac{375 \sin 15^\circ}{\sin 72^\circ} \approx 102.1}$

41 · Practice · KBZ eSolutions manual · PDF p.100 · answer A

In a triangle, ∠A = 50°, ∠C = 68° and the side opposite C is 5. Find the side x opposite A to the nearest tenth.

1. $\frac{\sin 50^\circ}{x} = \frac{\sin 68^\circ}{5}$ law of sines
2. $\boxed{x = \frac{5 \sin 50^\circ}{\sin 68^\circ} \approx 4.1}$